KITABU QUEST

S1

MATHEMATICS

Grey Crowned Crane

Cover scene

learners solving real-life mathematics with measuring tools and a grey crowned crane mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE 1

UGA NCDC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST S1 Mathematics

Uganda NCDC learner book

Mascot: grey crowned crane

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Uganda learners.

How To Use This Book

This book helps S1 learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

The Uganda Vision 2040 aims to transform Uganda into a modern: Lesson Opener

Learning focus: The Uganda Vision 2040 aims to transform Uganda into a modern

Country link: a Uganda school, market, farming, building, transport, or data problem for The Uganda Vision 2040 aims to transform Uganda into a modern.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about the uganda vision 2040 aims to transform uganda into a modern

Inquiry questions:
- How can the uganda vision 2040 aims to transform uganda into a modern help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

The Uganda Vision 2040 aims to transform Uganda into a modern: Learn

The Uganda Vision 2040 aims to transform Uganda into a modern teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- The Uganda Vision 2040 aims to transform Uganda into a modern
- Uganda
- Vision
- 2040
- aims
- transform
- into
- modern

The Uganda Vision 2040 aims to transform Uganda into a modern: Worked Example

Worked example for The Uganda Vision 2040 aims to transform Uganda into a modern: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Uganda classroom, market, garden, transport, or home example.

The Uganda Vision 2040 aims to transform Uganda into a modern: Vocabulary And Study Skill

Use these words and study moves before you practise The Uganda Vision 2040 aims to transform Uganda into a modern. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- The Uganda Vision 2040 aims to transform Uganda into a modern
- Uganda
- Vision
- 2040
- aims
- transform
- into
- modern

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Uganda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about The Uganda Vision 2040 aims to transform Uganda into a modern without a clear example, method, or evidence.

The Uganda Vision 2040 aims to transform Uganda into a modern: Guided Activity

Guided activity for The Uganda Vision 2040 aims to transform Uganda into a modern:

1. Work with a partner and read the worked example.
2. Make a similar question using Uganda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

The Uganda Vision 2040 aims to transform Uganda into a modern: Practice And Correction

Practice for The Uganda Vision 2040 aims to transform Uganda into a modern:

Main outcome: solve and explain problems about the uganda vision 2040 aims to transform uganda into a modern.

1. Solve one the uganda vision 2040 aims to transform uganda into a modern question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Uganda home, school, shop, garden, road, game, or timetable example connected to the uganda vision 2040 aims to transform uganda into a modern and solve it.

The Uganda Vision 2040 aims to transform Uganda into a modern: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from The Uganda Vision 2040 aims to transform Uganda into a modern.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

The Uganda Vision 2040 aims to transform Uganda into a modern: Outcome Check

Outcome check for The Uganda Vision 2040 aims to transform Uganda into a modern: Solve and explain problems about the uganda vision 2040 aims to transform uganda into a modern.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

KEY CHANGES IN THE CURRICULUM: Lesson Opener

Learning focus: KEY CHANGES IN THE CURRICULUM

Country link: a Uganda school, market, farming, building, transport, or data problem for KEY CHANGES IN THE CURRICULUM.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about key changes in the curriculum

Inquiry questions:
- How can key changes in the curriculum help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

KEY CHANGES IN THE CURRICULUM: Learn

KEY CHANGES IN THE CURRICULUM teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- KEY CHANGES IN THE CURRICULUM
- CHANGES
- CURRICULUM
- method
- model
- unit
- answer
- check

KEY CHANGES IN THE CURRICULUM: Worked Example

Worked example for KEY CHANGES IN THE CURRICULUM: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Uganda classroom, market, garden, transport, or home example.

KEY CHANGES IN THE CURRICULUM: Vocabulary And Study Skill

Use these words and study moves before you practise KEY CHANGES IN THE CURRICULUM. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- KEY CHANGES IN THE CURRICULUM
- CHANGES
- CURRICULUM
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Uganda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about KEY CHANGES IN THE CURRICULUM without a clear example, method, or evidence.

KEY CHANGES IN THE CURRICULUM: Guided Activity

Guided activity for KEY CHANGES IN THE CURRICULUM:

1. Work with a partner and read the worked example.
2. Make a similar question using Uganda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

KEY CHANGES IN THE CURRICULUM: Practice And Correction

Practice for KEY CHANGES IN THE CURRICULUM:

Main outcome: solve and explain problems about key changes in the curriculum.

1. Solve one key changes in the curriculum question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Uganda home, school, shop, garden, road, game, or timetable example connected to key changes in the curriculum and solve it.

KEY CHANGES IN THE CURRICULUM: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from KEY CHANGES IN THE CURRICULUM.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

KEY CHANGES IN THE CURRICULUM: Outcome Check

Outcome check for KEY CHANGES IN THE CURRICULUM: Solve and explain problems about key changes in the curriculum.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

THE: Lesson Opener

Learning focus: THE

Country link: a Uganda school, market, farming, building, transport, or data problem for THE.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about the

Inquiry questions:
- How can the help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

THE: Learn

THE teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- THE
- method
- model
- unit
- answer
- check

THE: Worked Example

Worked example for THE: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Uganda classroom, market, garden, transport, or home example.

THE: Vocabulary And Study Skill

Use these words and study moves before you practise THE. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- THE
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Uganda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about THE without a clear example, method, or evidence.

THE: Guided Activity

Guided activity for THE:

1. Work with a partner and read the worked example.
2. Make a similar question using Uganda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

THE: Practice And Correction

Practice for THE:

Main outcome: solve and explain problems about the.

1. Solve one the question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Uganda home, school, shop, garden, road, game, or timetable example connected to the and solve it.

THE: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from THE.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

THE: Outcome Check

Outcome check for THE: Solve and explain problems about the.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

THE CURRICULUM: Lesson Opener

Learning focus: THE CURRICULUM

Country link: a Uganda school, market, farming, building, transport, or data problem for THE CURRICULUM.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about the curriculum

Inquiry questions:
- How can the curriculum help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

THE CURRICULUM: Learn

THE CURRICULUM teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- THE CURRICULUM
- CURRICULUM
- method
- model
- unit
- answer
- check

THE CURRICULUM: Worked Example

Worked example for THE CURRICULUM: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Uganda classroom, market, garden, transport, or home example.

THE CURRICULUM: Vocabulary And Study Skill

Use these words and study moves before you practise THE CURRICULUM. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- THE CURRICULUM
- CURRICULUM
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Uganda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about THE CURRICULUM without a clear example, method, or evidence.

THE CURRICULUM: Guided Activity

Guided activity for THE CURRICULUM:

1. Work with a partner and read the worked example.
2. Make a similar question using Uganda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

THE CURRICULUM: Practice And Correction

Practice for THE CURRICULUM:

Main outcome: solve and explain problems about the curriculum.

1. Solve one the curriculum question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Uganda home, school, shop, garden, road, game, or timetable example connected to the curriculum and solve it.

THE CURRICULUM: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from THE CURRICULUM.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

THE CURRICULUM: Outcome Check

Outcome check for THE CURRICULUM: Solve and explain problems about the curriculum.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Generic Skills: Lesson Opener

Learning focus: Generic Skills

Country link: a Uganda school, market, farming, building, transport, or data problem for Generic Skills.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about generic skills

Inquiry questions:
- How can generic skills help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Generic Skills: Learn

Generic Skills teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Generic Skills
- Generic
- Skills
- method
- model
- unit
- answer
- check

Generic Skills: Worked Example

Worked example for Generic Skills: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Uganda classroom, market, garden, transport, or home example.

Generic Skills: Vocabulary And Study Skill

Use these words and study moves before you practise Generic Skills. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Generic Skills
- Generic
- Skills
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Uganda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Generic Skills without a clear example, method, or evidence.

Generic Skills: Guided Activity

Guided activity for Generic Skills:

1. Work with a partner and read the worked example.
2. Make a similar question using Uganda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Generic Skills: Practice And Correction

Practice for Generic Skills:

Main outcome: solve and explain problems about generic skills.

1. Solve one generic skills question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Uganda home, school, shop, garden, road, game, or timetable example connected to generic skills and solve it.

Generic Skills: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Generic Skills.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Generic Skills: Outcome Check

Outcome check for Generic Skills: Solve and explain problems about generic skills.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

The Uganda Vision 2040 aims to transform Uganda into a modern: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from The Uganda Vision 2040 aims to transform Uganda into a modern that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Uganda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

KEY CHANGES IN THE CURRICULUM: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from KEY CHANGES IN THE CURRICULUM that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Uganda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

THE: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from THE that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Uganda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

THE CURRICULUM: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from THE CURRICULUM that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Uganda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

The Uganda Vision 2040 aims to transform Uganda into a modern: Application Workshop 1

Application workshop 1 for The Uganda Vision 2040 aims to transform Uganda into a modern.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

KEY CHANGES IN THE CURRICULUM: Application Workshop 2

Application workshop 2 for KEY CHANGES IN THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE: Application Workshop 3

Application workshop 3 for THE.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE CURRICULUM: Application Workshop 4

Application workshop 4 for THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Generic Skills: Application Workshop 5

Application workshop 5 for Generic Skills.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

The Uganda Vision 2040 aims to transform Uganda into a modern: Application Workshop 6

Application workshop 6 for The Uganda Vision 2040 aims to transform Uganda into a modern.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

KEY CHANGES IN THE CURRICULUM: Application Workshop 7

Application workshop 7 for KEY CHANGES IN THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE: Application Workshop 8

Application workshop 8 for THE.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE CURRICULUM: Application Workshop 9

Application workshop 9 for THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Generic Skills: Application Workshop 10

Application workshop 10 for Generic Skills.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

The Uganda Vision 2040 aims to transform Uganda into a modern: Application Workshop 11

Application workshop 11 for The Uganda Vision 2040 aims to transform Uganda into a modern.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

KEY CHANGES IN THE CURRICULUM: Application Workshop 12

Application workshop 12 for KEY CHANGES IN THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE: Application Workshop 13

Application workshop 13 for THE.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE CURRICULUM: Application Workshop 14

Application workshop 14 for THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Generic Skills: Application Workshop 15

Application workshop 15 for Generic Skills.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

The Uganda Vision 2040 aims to transform Uganda into a modern: Application Workshop 16

Application workshop 16 for The Uganda Vision 2040 aims to transform Uganda into a modern.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

KEY CHANGES IN THE CURRICULUM: Application Workshop 17

Application workshop 17 for KEY CHANGES IN THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE: Application Workshop 18

Application workshop 18 for THE.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE CURRICULUM: Application Workshop 19

Application workshop 19 for THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Generic Skills: Application Workshop 20

Application workshop 20 for Generic Skills.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

The Uganda Vision 2040 aims to transform Uganda into a modern: Application Workshop 21

Application workshop 21 for The Uganda Vision 2040 aims to transform Uganda into a modern.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

KEY CHANGES IN THE CURRICULUM: Application Workshop 22

Application workshop 22 for KEY CHANGES IN THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE: Application Workshop 23

Application workshop 23 for THE.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE CURRICULUM: Application Workshop 24

Application workshop 24 for THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Generic Skills: Application Workshop 25

Application workshop 25 for Generic Skills.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

The Uganda Vision 2040 aims to transform Uganda into a modern: Application Workshop 26

Application workshop 26 for The Uganda Vision 2040 aims to transform Uganda into a modern.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

KEY CHANGES IN THE CURRICULUM: Application Workshop 27

Application workshop 27 for KEY CHANGES IN THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE: Application Workshop 28

Application workshop 28 for THE.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE CURRICULUM: Application Workshop 29

Application workshop 29 for THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Generic Skills: Application Workshop 30

Application workshop 30 for Generic Skills.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

The Uganda Vision 2040 aims to transform Uganda into a modern: Application Workshop 31

Application workshop 31 for The Uganda Vision 2040 aims to transform Uganda into a modern.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

KEY CHANGES IN THE CURRICULUM: Application Workshop 32

Application workshop 32 for KEY CHANGES IN THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE: Application Workshop 33

Application workshop 33 for THE.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

THE CURRICULUM: Application Workshop 34

Application workshop 34 for THE CURRICULUM.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Uganda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Glossary

Use this glossary to revise important words in Mathematics.

- The Uganda Vision 2040 aims to transform Uganda into a modern: Explain this term using an example from the book, your school, or your community.
- KEY CHANGES IN THE CURRICULUM: Explain this term using an example from the book, your school, or your community.
- THE: Explain this term using an example from the book, your school, or your community.
- THE CURRICULUM: Explain this term using an example from the book, your school, or your community.
- Generic Skills: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.